

TMFC038 Resource Performance Management 1.2.0

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Notice

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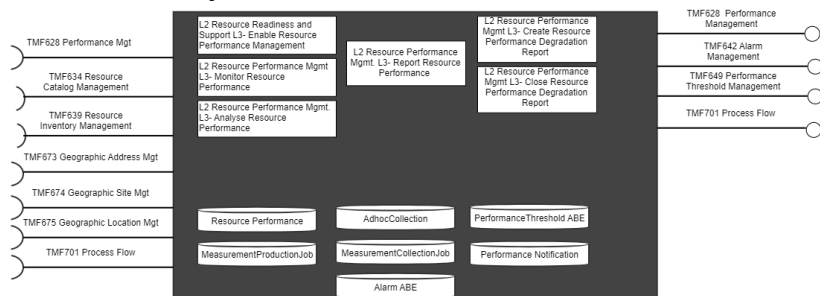
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1. Overview

1. [TAC-208](#) IG1171 (update) Component Definition to v4.0.0 and incorporate IG1245 Principles to Define ODA Components
2. [\[TAC-250\] IG 1171 Improvements Some observations & recommendations. - TM Forum JIRA](#)
3. [\[TAC-214\] Interface Standardization needs all 3 stages of process to be developed - TM Forum JIRA](#)
4. [\[TAC-226\] Overview - TM Forum JIRA](#)
5. [ODA-846](#) Summary of ODA component Template enhancements for 14th Sep Review

Component Name	ID	Description	ODA Function Block
<i>Resource Performance Management</i>	TMFC 038	Resource Performance Management will collect, monitor, analyze, and report on the performance of the service provider's resources	Intelligence Management

TMFC038 : Resource Performance Management



LEGEND

eTOM Business Activities

SID Data Entities

Dependent APIs

Exposed APIs

2. eTOM Processes, SID Data Entities and Functional Framework Functions

2.1. eTOM business activities

<Note to not be inserted onto ODA Component specifications: If a new ABE is required, but it does not yet exist in SID. Then you should include a textual description of the new ABE, and it should be clearly noted that this ABE does not yet exist. In addition a Jira epic should be raised to request the new ABE is added to SID, and the SID team should be consulted. Finally, a decision is required on the feasibility of the component without this ABE. If the ABE is critical then the component specification should not be published until the ABE issue has been resolved. Alternatively if the ABE is not critical, then the specification could continue to publication. The result of this decision should be clearly recorded.>

eTOM business activities this ODA Component is responsible for.

Identifier	Level	Business Activity Name	Description
1.5.4	L2	Resource Readiness and Support	Resource Readiness & Support processes are responsible for managing resource infrastructure to ensure that appropriate application, computing and network resources are available and ready to support the Fulfilment, Assurance and Billing processes in instantiating and managing resource instances, and for monitoring and reporting on the capabilities and costs of the individual FAB processes. Responsibilities of these processes include but are not limited to: • Supporting the operational introduction of new and/or modified resource infrastructure and conducting operations readiness testing and acceptance; • Managing planned outages; • Managing and ensuring the ongoing quality of the Resource Inventory; • Analyzing availability and performance over time on resources or groups of resources, including trend analysis and forecasting; • Demand balancing in order to maintain resource capacity and performance • Performing proactive maintenance and repair activities; • Establishing and managing the workforce to support the eTOM processes • Managing spares, repairs, warehousing, transport and distribution of resources and consumable goods. • Conducting Vulnerability Management; • Conducting Threat Assessments; • Conducting Risk Assessments; • Conducting Risk Mitigation; • Conducting Secure Configuration Activities
1.5.4.2	L3	Enable Resource Performance Management	The responsibilities of the Enable Resource Performance Management processes are twofold - support Resource Performance Management processes by proactively monitoring and assessing resource infrastructure performance, and monitoring, managing and reporting on the capability of the Resource Performance Management processes. Proactive management is undertaken using a range of performance parameters, whether technical, time, economic or process related. The responsibilities of the processes include, but are not limited to: • Undertaking proactive monitoring regimes of resource infrastructure as required to ensure ongoing performance within agreed parameters over time; • Developing and maintaining a repository of acceptable performance threshold standards for resource instances to support the Resource Performance Management processes; • Undertaking trend analysis, and producing reports, of the performance of resource infrastructure to identify any longer term deterioration; • Monitoring and analyzing the resource instance analyses produced by the Resource Performance Management processes to identify problems that may be applicable to the resource infrastructure as a whole; • Sourcing details relating to resource instance performance and analysis from the resource inventory to assist in the development of trend analyses; • Logging the results of the analysis into the resource inventory repository; • Establishing and managing resource performance data collection schedules, including managing the collection of the necessary information from the Resource Data Collection & Distribution processes, to support proactive monitoring and analysis activity, and requests from Resource Performance Management processes for additional data to support resource instance performance analysis; • Establishing and managing facilities to support management of planned resource infrastructure and resource instance outages; • Establishing, maintaining and managing the testing of resource performance control plans to cater for anticipated resource performance disruptions; • Proactively triggering the instantiation of control plans to manage performance through programmed and/or foreseen potentially disruptive events, i.e. anticipated traffic loads on Xmas day, planned outages, etc.; • Tracking and monitoring of the Resource Performance Management processes and associated costs (including where resource infrastructure is deployed and managed by third parties), and reporting on the capability of the Resource Performance Management processes; • Establishing and managing resource performance notification facilities and lists to support the Resource Performance Management notification and reporting processes • Supporting the Support Service Quality Management process.
1.5.9	L2	Resource Performance Management	Resource Performance Management processes encompass managing, tracking, monitoring, analyzing, controlling and reporting on the performance of specific resources. They work with basic information received from the Resource Data Collection & Distribution processes. If the analysis identifies a resource performance violation or a potential service performance violation, information will be passed to Resource Trouble Management and/or Service Quality Management as appropriate. The latter processes are responsible for deciding on and carrying out the appropriate action/response. This may include requests to the Resource Performance Management processes to install controls to optimize the specific resource performance. The Resource Performance Management processes will continue to track the resource performance problem, ensuring that resource performance is restored to a level required to support services.
1.5.9.1	L3	Monitor Resource Performance	The objective of the Monitor Resource Performance processes is to monitor received resource performance information and undertake first-in detection. The responsibilities of the processes include, but are not limited to: •Undertaking the role of first in detection by monitoring the received specific resource performance data; •Comparing the received specific resource performance data to performance standards set for each specific resource (available from the Resource Inventory); •Assessing and recording received specific resource performance data which is within tolerance limits for performance standards, and for which continuous monitoring and measuring of specific resource performance is required; •Recording the results of the continuous monitoring for reporting through the Report Resource Performance processes; •Detecting performance threshold violations which represent specific resource failures due to abnormal performance; •Passing information about resource failures due to performance threshold violations to Resource Trouble Management to manage any necessary restoration activity as determined by that process; •Passing information about potential specific service performance degradations arising from specific resource degradations to Service Quality Management to manage any necessary restoration activity as determined by that process; •Detecting performance degradation for specific resources which provide early warning of potential issues; •Forwarding resource performance degradation notifications to other Resource Performance Management processes, which manage activities to restore normal specific resource performance •Logging specific resource performance degradation and violation details within the repository in the Manage Resource Inventory processes to ensure historical records are available to support the needs of other processes.
1.5.9.2	3	Analyze Resource Performance	The objective of the Analyze Resource Performance processes is to analyze the information received from the Monitor Resource Performance process to evaluate the performance of a specific resource. The responsibilities of the processes include, but are not limited to: •Undertaking analysis as required on specific resource performance information received from the Monitor Resource Performance processes; •Initiating, modifying and cancelling continuous performance data collection schedules for specific resources required to analyze specific resource performance. These schedules are established through requests sent to the Enable Resource Data Collection & Distribution processes; •Determining the root causes of specific resource performance degradations and violations; •Recording the results of the analysis and intermediate updates in the Resource Inventory for historical analysis and for use as required by other processes •Undertaking specific detailed analysis (if the original requested came from Service Quality Management processes) to discover the root cause of service performance degradations that may be arising due to interactions between resource instances, without any specific resource instance having an unacceptable performance in its own right.
1.5.9.4	3	Report Resource Performance	The objective of the Report Resource Performance processes is to monitor the status of resource performance degradation reports, provide notifications of any changes and provide management reports. These processes are responsible for continuously monitoring the status of resource performance degradation reports and managing notifications to other processes in the Resource-Ops and other layers, and to other parties registered to receive notifications of any status changes. Notification lists are managed and maintained by the Enable Resource Performance Management processes. These processes record, analyze and assess the resource performance degradation report status changes to provide management reports and any specialized summaries of the efficiency and effectiveness of the overall Resource Performance Management process. These specialized summaries could be specific reports required by specific audiences.
1.5.9.5	3	Create Resource Performance Degradation Report	The objective of the Create Resource Performance Degradation Report process is to create a new resource performance degradation report, modify existing resource performance degradation reports, and request cancellation of existing resource performance degradation reports. A new resource performance degradation report may be created as a result of specific resource performance notifications undertaken by the Monitor Resource Performance processes, or at the request of analysis undertaken by other Service-Ops or Resource-Ops or party management processes which detect that some form of deterioration or failure has occurred requires an assessment of the specific resource performance. If the resource performance degradation report is created as a result of a notification or request from processes other than Monitor Resource Performance processes, the Create Resource Performance Degradation Report processes are responsible for converting the received information into a form suitable for the Resource Performance Management processes, and for requesting additional information if required.

1.5.9.7	3	Close Resource Performance Degradation Report	The objective of the Close Resource Performance Degradation Report processes is to close a resource performance degradation report when the resource performance has been resolved. These processes monitor the status of all open resource performance degradation reports, and recognize that a resource performance degradation report is ready to be closed when the status is changed to cleared.
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2.2. SID ABEs

<Note not to be inserted into ODA Component specifications: If a new ABE is required, but it does not yet exist in SID. Then you should include a textual description of the new ABE, and it should be clearly noted that this ABE does not yet exist. In addition a Jira epic should be raised to request the new ABE is added to SID, and the SID team should be consulted. Finally, a decision is required on the feasibility of the component without this ABE. If the ABE is critical then the component specification should not be published until the ABE issue has been resolved. Alternatively if the ABE is not critical, then the specification could continue to publication. The result of this decision should be clearly recorded.>

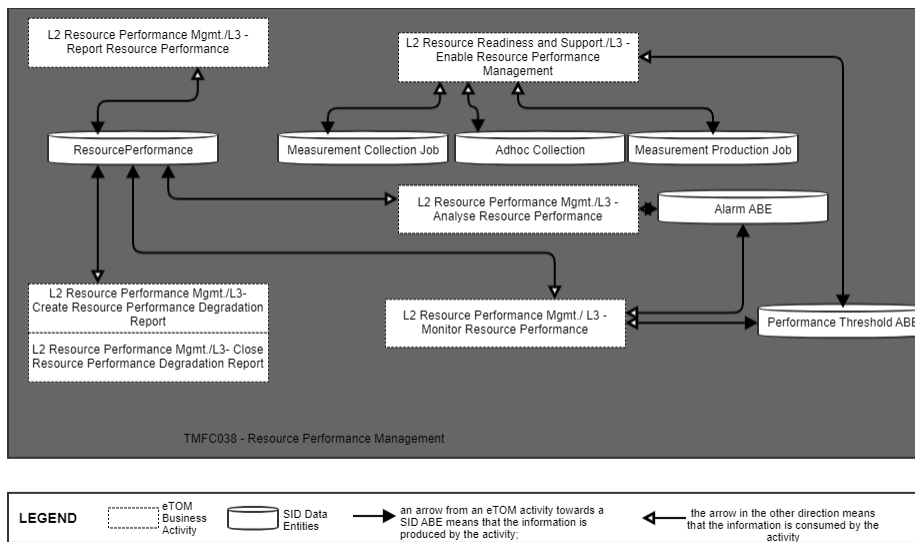
SID ABEs this ODA Component is responsible for:

SID ABE Level 1	SID ABE Level 2 (or set of BEs)*
Resource Performance ABE	Resource Performance
Performance ABE	MeasurementProductionJob
Performance ABE	AdhocCollection
	MeasurementCollectionJob
Performance ABE	Performance Threshold ABE
Resource Trouble ABE	Alarm ABE

*: if SID ABE Level 2 is not specified this means that all the L2 business entities must be implemented, else the L2 SID ABE Level is specified.

2.3. eTOM L2 - SID ABEs links

eTOM L2 vs SID ABEs links for this ODA Component.



2.4. Functional Framework Functions

Function ID	Function Name	Function Description	Aggregate Function Level 1	Aggregate Function Level 2
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498	Resource Performance Data Analyzing	Resource Performance Data Analyzing functions provide the necessary functionality to analyze the performance of the various service provider's resources. This includes: Analyzing performance data received from Resource Performance Monitoring and historic performance data or system environment data • Notify operations applications for automatic or manual action in case of alarming analysis results. • Determining the root causes of resource performance degradations	Resource Performance Management	Resource Performance Supervision
499	Resource Performance Data Aggregation and Trend Analyzing	Resource Performance Data Aggregation and Trend Analyzing provides data aggregation and trend analysis in resource performance monitoring	Resource Performance Management	Resource Performance Supervision
501	Resource Performance Event Correlation	Resource Performance Event Correlation provides performance event correlation in resource performance monitoring.	Resource Performance Management	Resource Performance Supervision
502	Resource Performance Data Accumulation	Resource Performance Data Accumulation provides performance data accumulation for resource performance monitoring, including real time monitoring data	Resource Performance Management	Resource Performance Supervision
903	Anomaly Monitoring	Anomaly Monitoring; System and network monitoring activity, determining whether it is normal or anomalous, based on rules, signatures or heuristics.	Resource Performance Management	Resource Performance Supervision
1066	Resource Performance Event Filtering	Resource Performance Data Accumulation provides performance data accumulation for resource performance monitoring, including real time monitoring data	Resource Performance Management	Resource Performance Supervision

3. TM Forum Open APIs & Events

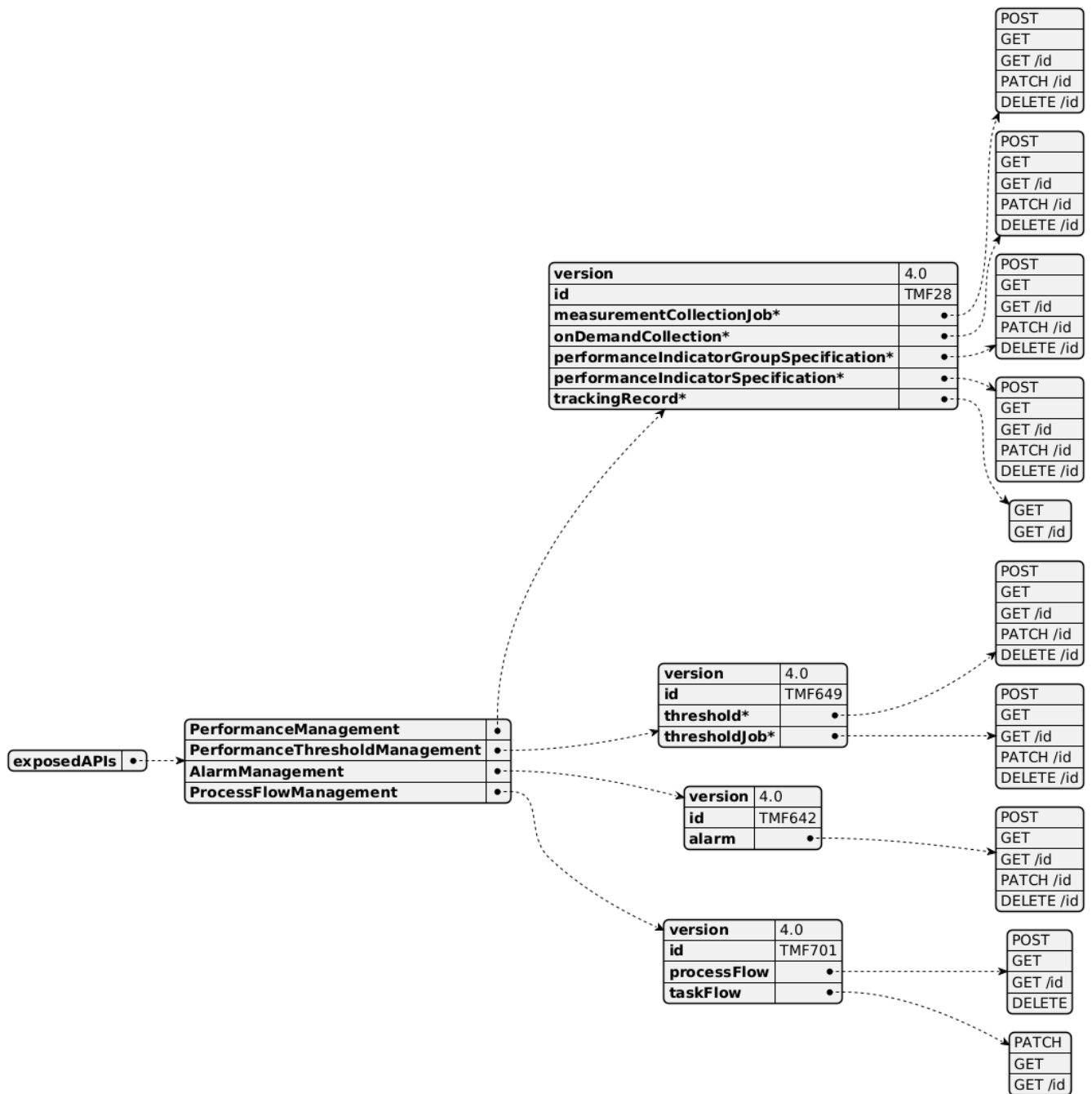
The following part covers the APIs and Events; This part is split in 3:

- List of **Exposed APIs** - This is the list of APIs available from this component.
- List of **Dependent APIs** - In order to satisfy the provided API, the component could require the usage of this set of required APIs.
- List of **Events (generated & consumed)** - The events which the component may generate is listed in this section along with a list of the events which it may consume. Since there is a possibility of multiple sources and receivers for each defined event.

<Note note to be inserted into ODA Component specifications: If a new Open API is required, but it does not yet exist. Then you should include a textual description of the new Open API, and it should be clearly noted that this Open API does not yet exist. In addition, a Jira epic should be raised to request the new Open API is added, and the Open API team should be consulted. Finally, a decision is required on the feasibility of the component without this Open API. If the Open API is critical then the component specification should not be published until the Open API issue has been resolved. Alternatively if the Open API is not critical, then the specification could continue to publication. The result of this decision should be clearly recorded.>

3.1. Exposed APIs

Following diagram illustrates API/Resource/Operation:



API ID	API Name	API Version	Mandatory / Optional	Operations
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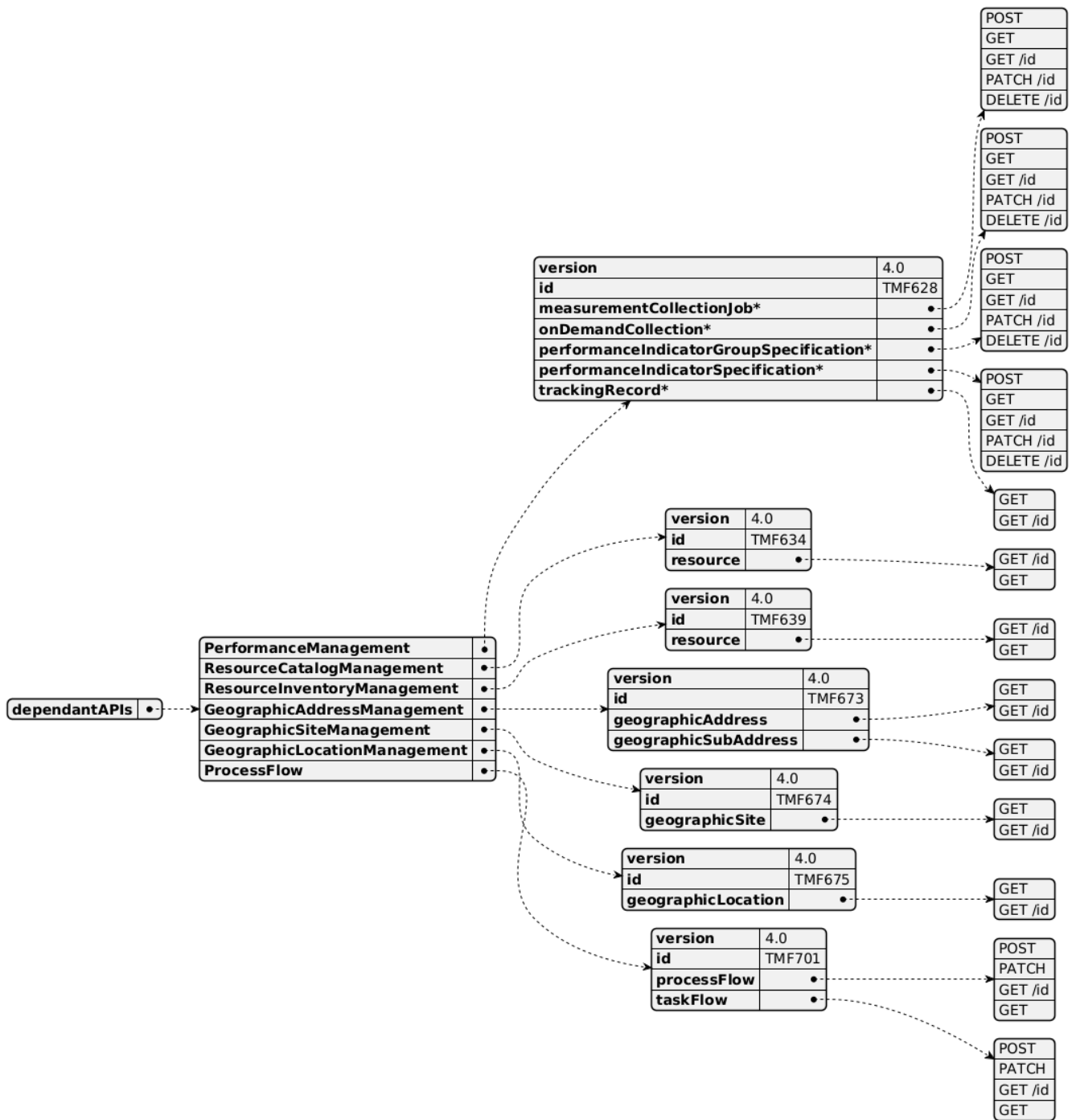
TMF628	TMF628 Performance Management API	4	Mandatory	measurementCollectionJob: - POST - GET - GET /id - PATCH /id - DELETE /id onDemandCollection: - POST - GET - GET /id - PATCH /id - DELETE /id performanceIndicatorGroupSpecification: - POST - GET - GET /id - PATCH /id - DELETE /id performanceIndicatorSpecification: - POST - GET - GET /id - PATCH /id - DELETE /id trackingRecord: - GET - GET /id
TMF642	TMF642 Alarm Management API	4	Optional	alarm: - POST - GET - GET /id - PATCH /id - DELETE /id
TMF649	TMF649 Performance Threshold Management API	4	Mandatory	threshold: - POST - GET - GET /id - PATCH /id - DELETE /id thresholdJob: - POST - GET - GET /id - PATCH /id - DELETE /id
TMF701	TMF701 Process Flow	4	Optional	processFlow: • GET • GET /id • POST • DELETE /id taskFlow: • GET • GET /id • PATCH /id

Note:

TMF 628 is in the beta version. TMF649 is in the historic table. Both of these need to be upgraded.

3.2. Dependent APIs

Following diagram illustrates API/Resource/Operation potentially used by the product catalog component:

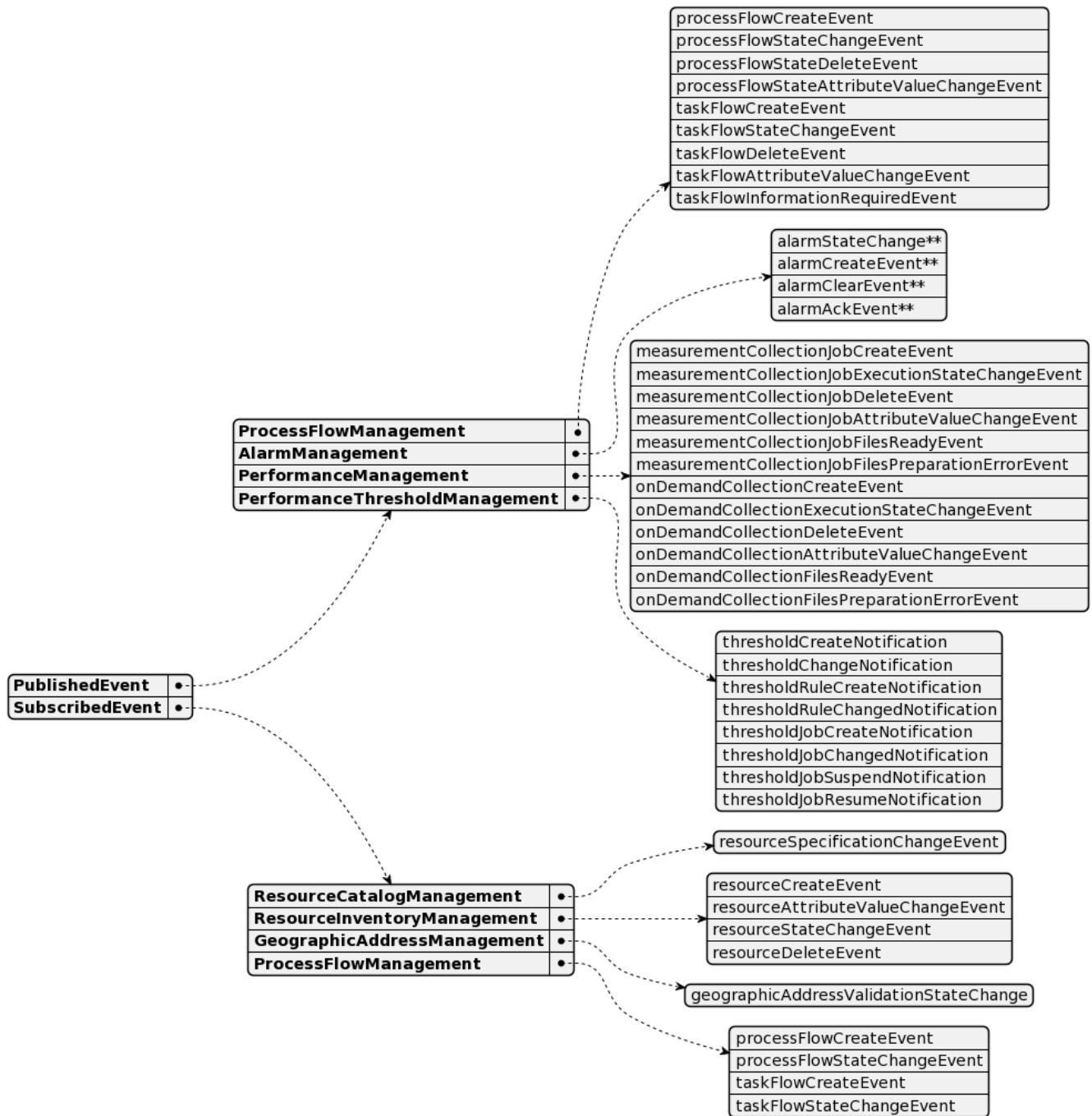


API ID	API Name	API Version	Mandatory / Optional	Operations	Rationales
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TMF628	TMF628 Performance Management API	4	Optional	measurementCollectionJob*: - POST - GET - GET /id - PATCH /id - DELETE /id onDemandCollection*: - POST - GET - GET /id - PATCH /id - DELETE /id performanceIndicatorGroupSpecification*: - POST - GET - GET /id - PATCH /id - DELETE /id performanceIndicatorSpecification*: - POST - GET - GET /id - PATCH /id - DELETE /id trackingRecord*: - GET - GET /id	
TMF634	TMF634 Resource Catalog Management API	4	Optional	resource: - GET /id - GET	
TMF639	TMF639 Resource Inventory Management API	4	Mandatory	resource: - GET /id - GET	Performance is collected, monitored and reported against a Resource instance that is stored in the Resource inventory.
TMF673	TMF673 Geographic Address Management	4	Optional	geographicAddress: - GET - GET /id geographicSubAddress: - GET - GET /id	
TMF674	TMF674 Geographic Site Management	4	Optional	geographicSite: - GET - GET /id	
TMF701	TMF701 Process Flow	4	Optional	processFlow: - POST - PATCH - GET /id - GET taskFlow: - POST - PATCH - GET /id - GET	

3.3. Events

The diagram illustrates the Events which the component may publish and the Events that the component may subscribe to and then may receive. Both lists are derived from the APIs listed in the preceding sections.



Note **: Alarm API documentation needs to be updated to include Alarm Create, Alarm Change, Alarm Clear, Alarm Ack

4. Machine Readable Component Specification

[TMForum-ODA-Ready-for-publication/1Beta2/TMFC038-ResourcePerformanceManagement/TMFC038-ResourcePerformanceManagement.yaml](https://github.com/tmforum-rand/TMForum-ODA-Ready-for-publication/blob/main/TMFC038-ResourcePerformanceManagement/TMFC038-ResourcePerformanceManagement.yaml) at main · tmforum-rand/TMForum-ODA-Ready-for-publication (github.com)

5. References

5.1. TMF Standards related versions

Standard	Version(s)
SID	24.0
eTOM	24.0
Functional Framework	24.0

5.2. Further resources

1. IG1228: please refer to IG1228 for defined use cases with ODA components interactions.

6. Administrative Appendix

6.1. Document History

6.1.1. Version History

Version Number	Date Modified	Modified by:	Description of changes
0.0.1	18 May 2023	Milind Bhagwat	First draft
1.0.0	13 Jun 2023	Amaia White	Finals edit prior to publication
1.0.1	25 Jul 2023	Ian Turkington	No content changed, simply a layout change to match template 3. Separated the YAML files to a managed repository.
1.0.1	14 Aug 2023	Amaia White	Final edits prior to publication
1.1.0	13 Aug 2024	Milind Bhagwat	Further edits
1.1.0	06 Sep 2024	Amaia White	Final edits prior to publication
1.2.0	17 Dec 2024	Milind Bhagwat	Removed Event API from exposed and dependent API list
1.2.0	27 Dec 2024	Rosie Wilson	Final edits prior to publication

6.1.2. Release History

Release Status	Date Modified	Modified by:	Description of changes
Pre-production	13-Jun-2023	Amaia White	Initial release.
Pre-production	14-Aug-2023	Amaia White	New release v1.0.1
Production	06-Oct-2023	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	06-Sept-2024	Amaia White	New release 1.1.0
Production	01-Nov-2024	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	18-Dec-2024	Rosie Wilson	New release 1.2.0
Production	07-Mar-2025	Adrienne Walcott	Updated to reflect TM Forum Approved status

6.2. Acknowledgements

This document was prepared by the members of the TM Forum Component and Canvas project team:

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Sylvie Demarest	Orange	Reviewer
Hugo Vaughan (TM Forum)	TM Forum	Additional Inputs
Ian Turkington	TM Forum	Additional Inputs

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